

3ART13 LED

Variant: CHECKED

2026-05-21

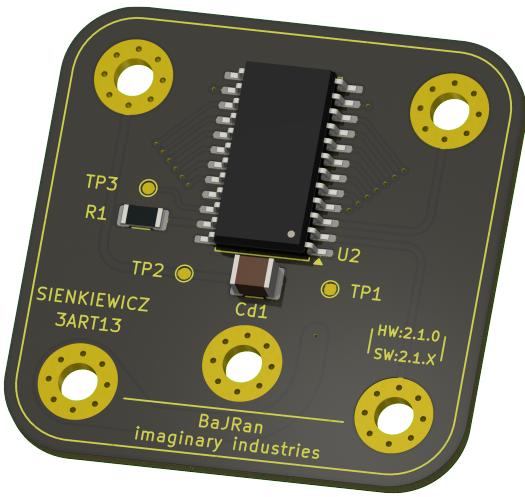
Rev 2.1.0+ (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 21-May-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

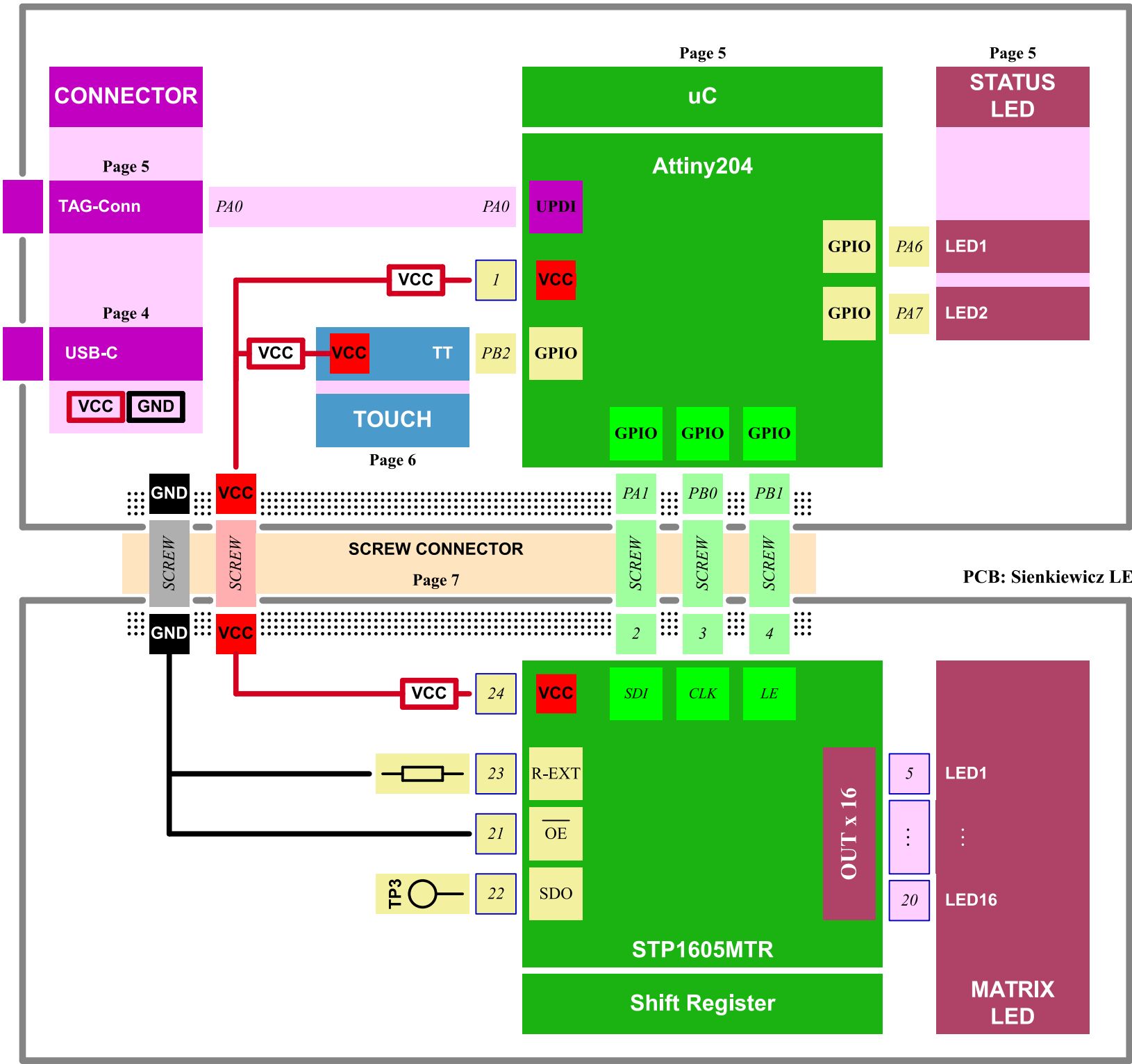
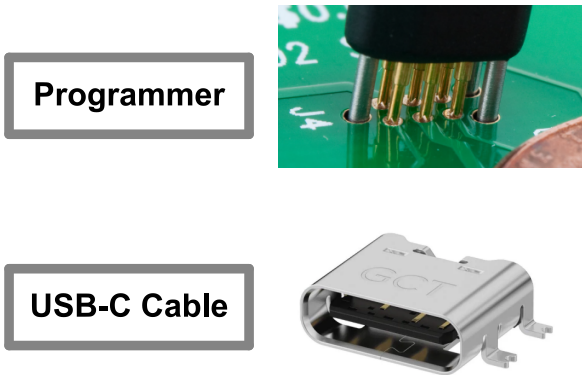
LAYOUT NOTE:

Example text for critical layout guidelines.

	Comments:	Company: BaJRan imaginary industries		Variant: CHECKED	Git Hash: 8dba9b7
	Sheet Title:	Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Path: /	File Name: Sienkiewicz_LED.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
			Reviewer:	Size: A3	Sheet: 1 of 8

[2] Block Diagram

PCB: Sienkiewicz TOUCH

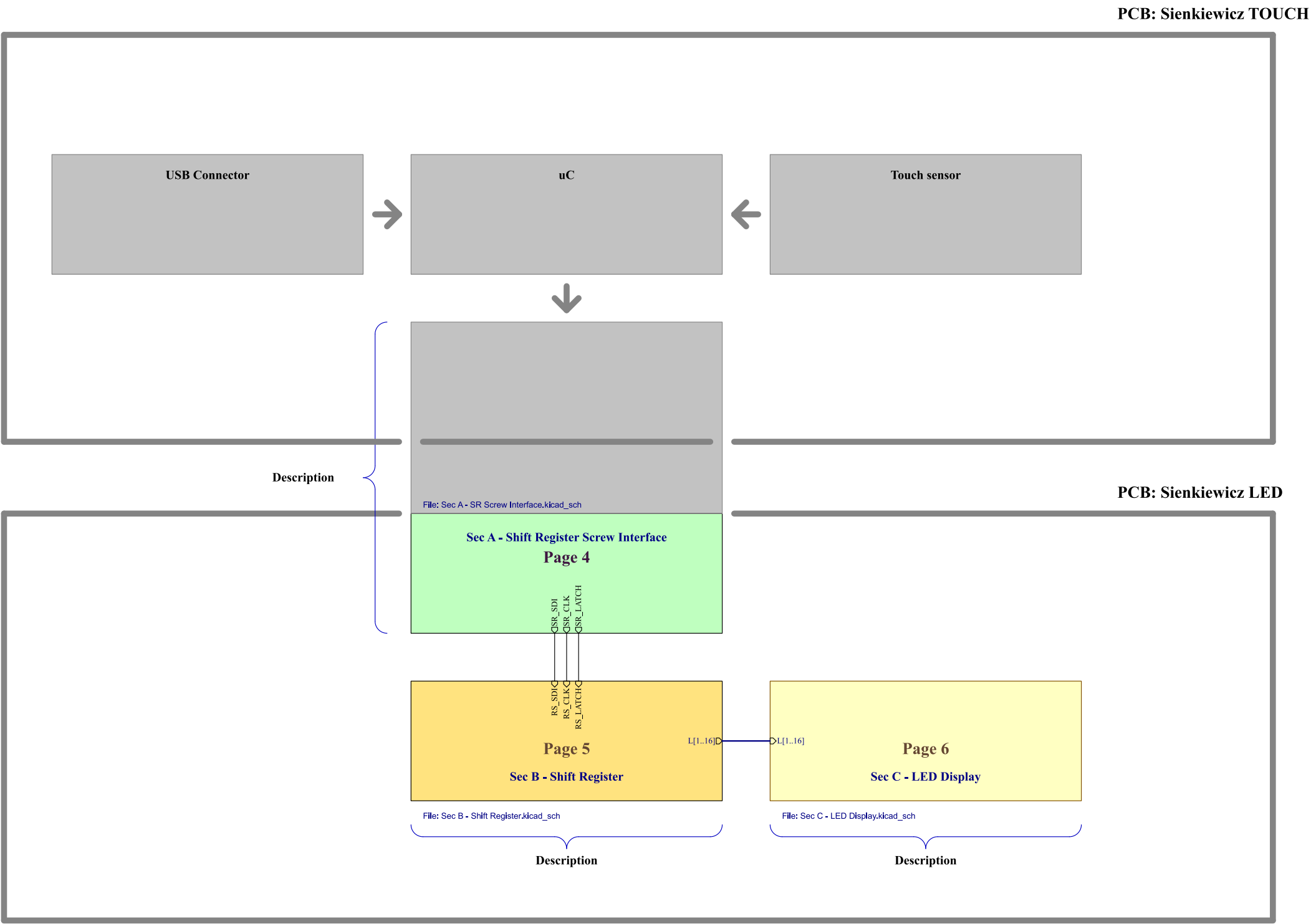


Target specifications:

Input voltage	5V
Current	200mA
Spec 3	----
Spec 4	----

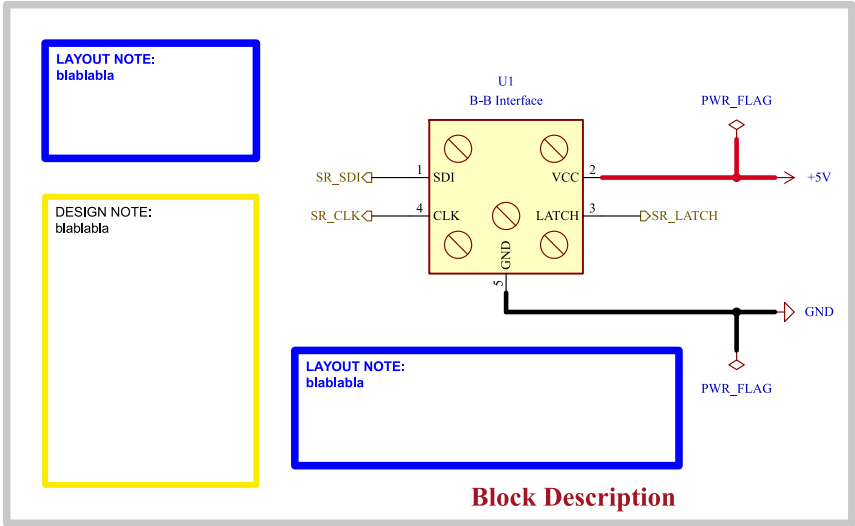
	Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 8dba9b7
	Sheet Title: Block Diagram	Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Path: /Block Diagram/	File Name: Block Diagram.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
			Reviewer:	Size: A3	Sheet: 2 of 8

[3] Project Architecture



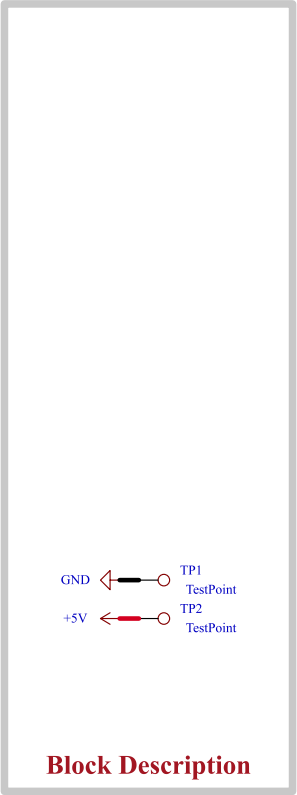
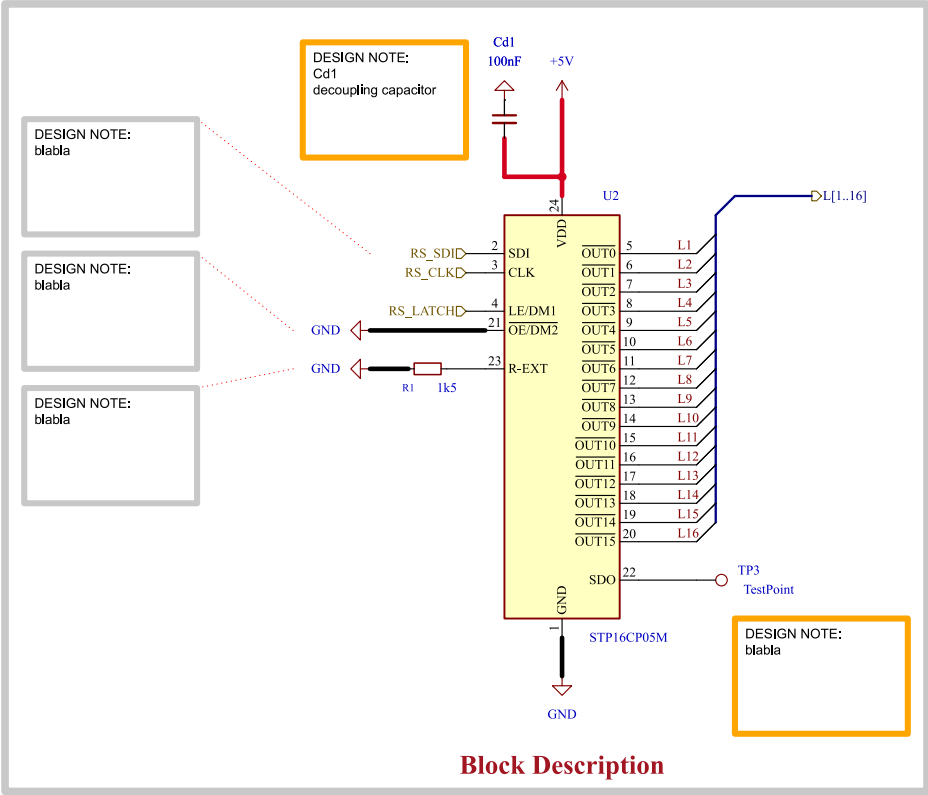
	Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 8dba9b7
	Sheet Title: Project Architecture	Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Path: /Project Architecture/	File Name: Project Architecture.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
			Reviewer:	Size: A3	Sheet: 3 of 8

[4] B-B Screw Interface



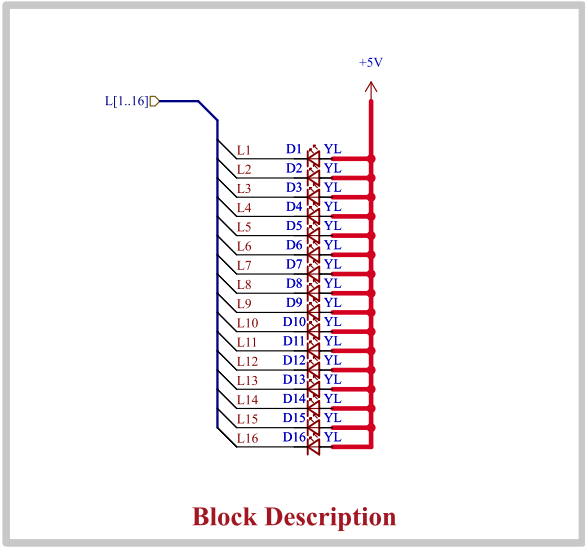
	Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	Company: BaJRanimgaginary industries		Variant: CHECKED	Git Hash: 8dba9b7
		Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Title: B-B Screw Interface	File Name: Sec A - SR Screw Interface.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
	Sheet Path: /Project Architecture/Sec A - Shift Register Screw Interface/		Reviewer:	Size: A4	Sheet: 4 of 8

[5] Shift Register



	Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 8dba9b7
		Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Title: Shift Register	File Name: Sec B - Shift Register.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
	Sheet Path: /Project Architecture/Sec B - Shift Register/		Reviewer:	Size: A4	Sheet: 5 of 8

[6] LED Display



Block Description

	Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 8dba9b7
		Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Title: LED Display	File Name: Sec C - LED Display.kicad_sch	Designer: Pawel Baran	Date: 2026-04-30	Revision: 2.1.0+ (Unreleased)
	Sheet Path: /Project Architecture/Sec C - LED Display/		Reviewer:	Size: A4	Sheet: 6 of 8

[7] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%

	Comments:	Company: BaJRan imaginary industries	Variant: CHECKED	Git Hash: 8dba9b7
		Board Name: 3ART13 LED	Project Name: Sienkiewicz	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12
	Revision: 2.1.0+ (Unreleased)			
Sheet Path: /Power - Sequencing/	Reviewer:	Size: A4	Sheet: 7 of 8	

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%

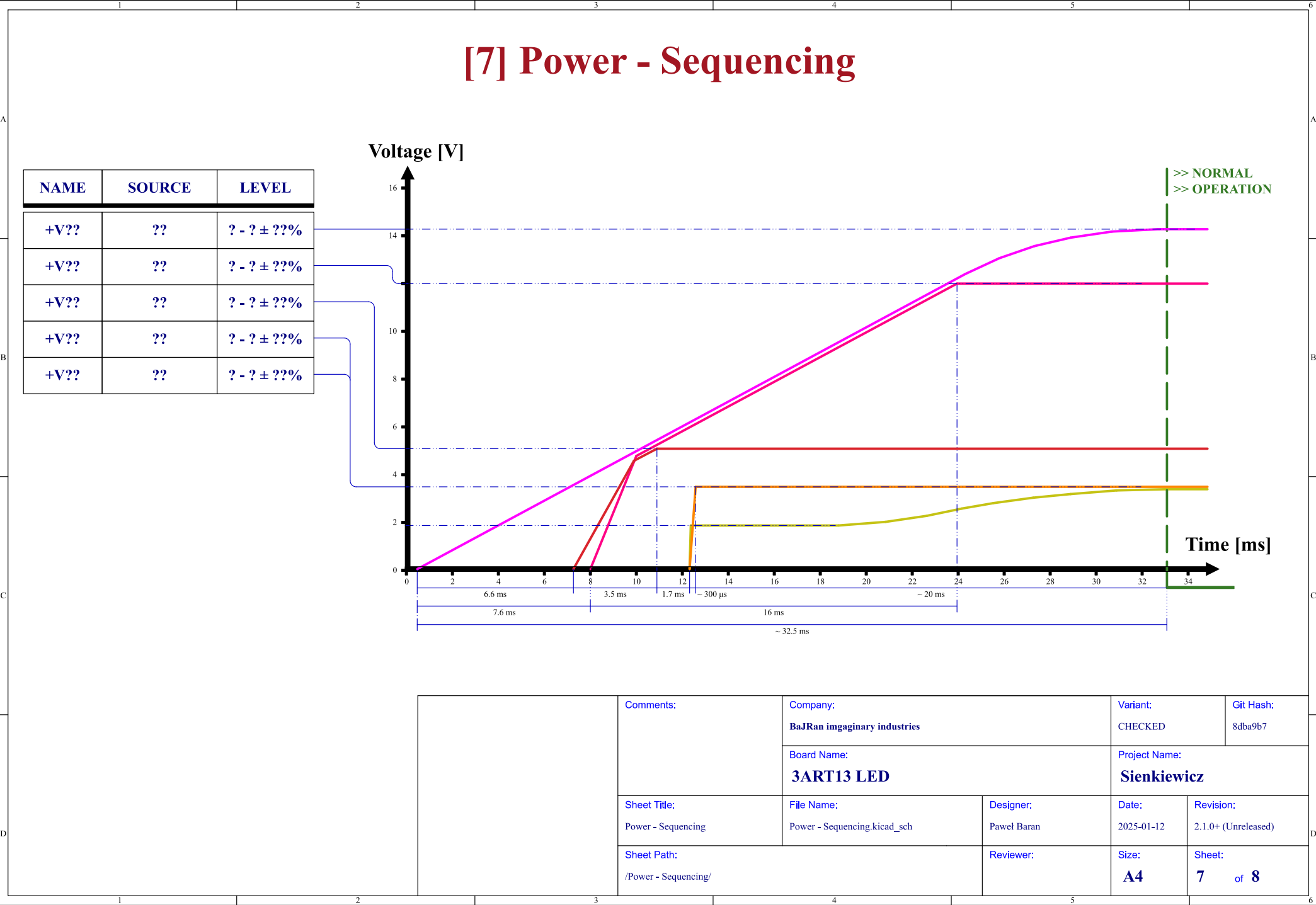
Voltage [V]

Time [ms]

>> NORMAL
>> OPERATION

6.6 ms, 7.6 ms, 3.5 ms, 1.7 ms, ~ 300 μs, 16 ms, ~ 20 ms, ~ 32.5 ms

	Comments:	Company: BaJRan imgaginary industries	Variant: CHECKED	Git Hash: 8dba9b7
		Board Name: 3ART13 LED	Project Name: Sienkiewicz	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12
	Sheet Path: /Power - Sequencing/		Reviewer:	Revision: 2.1.0+ (Unreleased)
			Size: A4	Sheet: 7 of 8



NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%

Voltage [V]

Time [ms]

>> NORMAL
>> OPERATION

6.6 ms
7.6 ms
3.5 ms
1.7 ms
~ 300 μs
16 ms
~ 20 ms
~ 32.5 ms

	Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 8dba9b7
		Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 7 of 8

[8] Revision History

Version 2.1.0 - 2026-05-02

- Added
- First version of PCB and Schematic

	Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: 8dba9b7
		Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 8 of 8